ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 1

Identify the Association Between Dependent and Independent Variables

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To identify and analyze the association between the dependent variable and independent variables using correlation analysis on the Diabetes dataset. The strength and direction of the linear relationship are measured using the Pearson correlation coefficient.

2. Objectives

- To load and examine the Diabetes dataset.
- To identify the dependent and independent variables.
- To calculate Pearson correlation coefficients between each independent variable and the target.
- To visualize relationships using a correlation bar chart, heatmap, and scatter plot.
- To identify the independent variable having the strongest linear association with the target.

3. Dataset Description

The Scikit-learn Diabetes dataset contains 442 observations and 10 independent variables. The dependent variable is the continuous target representing a quantitative measure of disease progression one year after baseline.

Property	Value
Number of samples	442
Independent variables	10
Dependent variable	target
Analysis technique	Pearson correlation

4. Theory

Correlation is a statistical technique used to measure the strength and direction of association between two variables. In this experiment, each independent feature is compared with the dependent variable (target).

The Pearson correlation coefficient, represented by r, ranges from -1 to +1. A value close to +1 indicates a strong positive linear relationship, a value close to -1 indicates a strong negative linear relationship, and a value close to 0 indicates a weak or negligible linear relationship.

The Pearson correlation coefficient can be expressed as:

$$\mathbf{r = Cov(X, Y) / (\sigma X \, \sigma Y)}$$

A positive correlation means that larger values of one variable tend to be associated with larger values of the other variable. A negative correlation means that larger values of one variable tend to be associated with smaller values of the other variable.

Correlation measures association, not causation. Therefore, a high correlation between two variables does not by itself prove that one variable causes changes in the other.

5. Methodology

- Load the Diabetes dataset using scikit-learn.
- Separate the independent variables from the target variable.
- Calculate Pearson correlation coefficients between each independent variable and target.
- Sort the correlation values to identify the strongest associations.
- Visualize the results using a bar chart and correlation heatmap.
- Plot BMI against the target to visually examine the strongest relationship.

6. Outputs

The following outputs were generated from the experiment.

6.1 Correlation Results

Independent Variable	Pearson Correlation
bmi	0.5865
s5	0.5659
bp	0.4415
s4	0.4305
s3	-0.3948
s6	0.3825
s1	0.2120
age	0.1879
s2	0.1741
sex	0.0431

6.2 Key Observation

The feature with the strongest absolute linear correlation with the target is 'bmi', with a Pearson correlation coefficient of 0.5865. This indicates a moderate positive linear association between BMI and the target.

6.3 Correlation Bar Chart

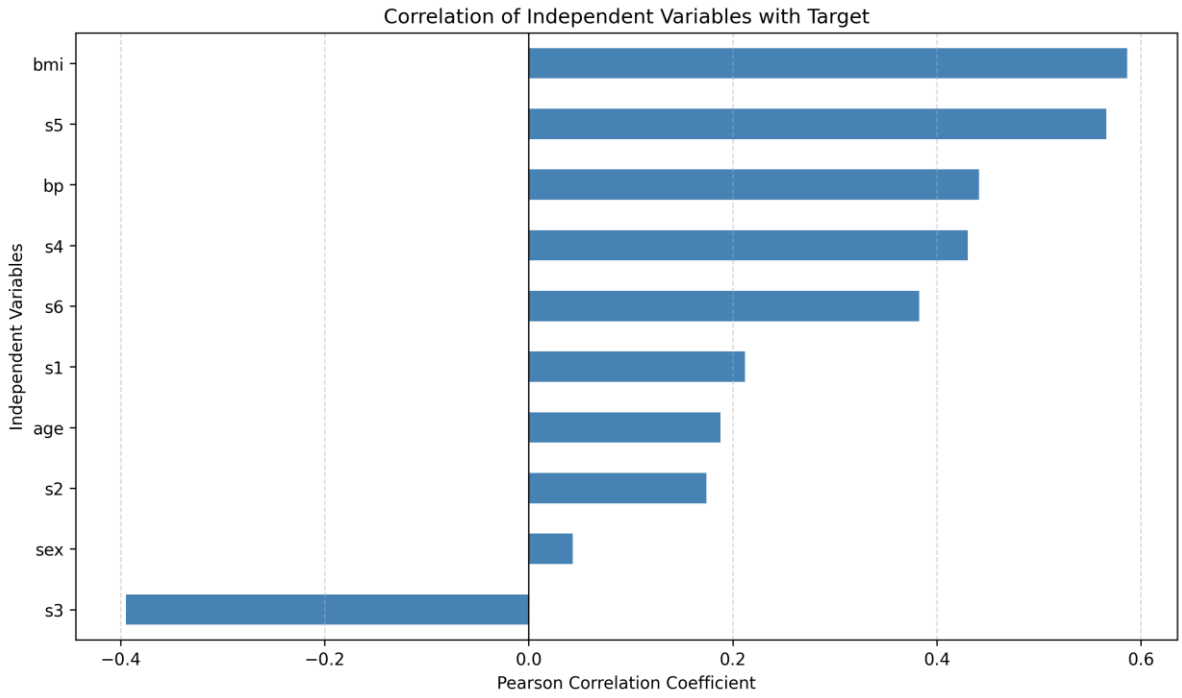


Figure 1: Pearson correlation of independent variables with the target

6.4 Correlation Heatmap

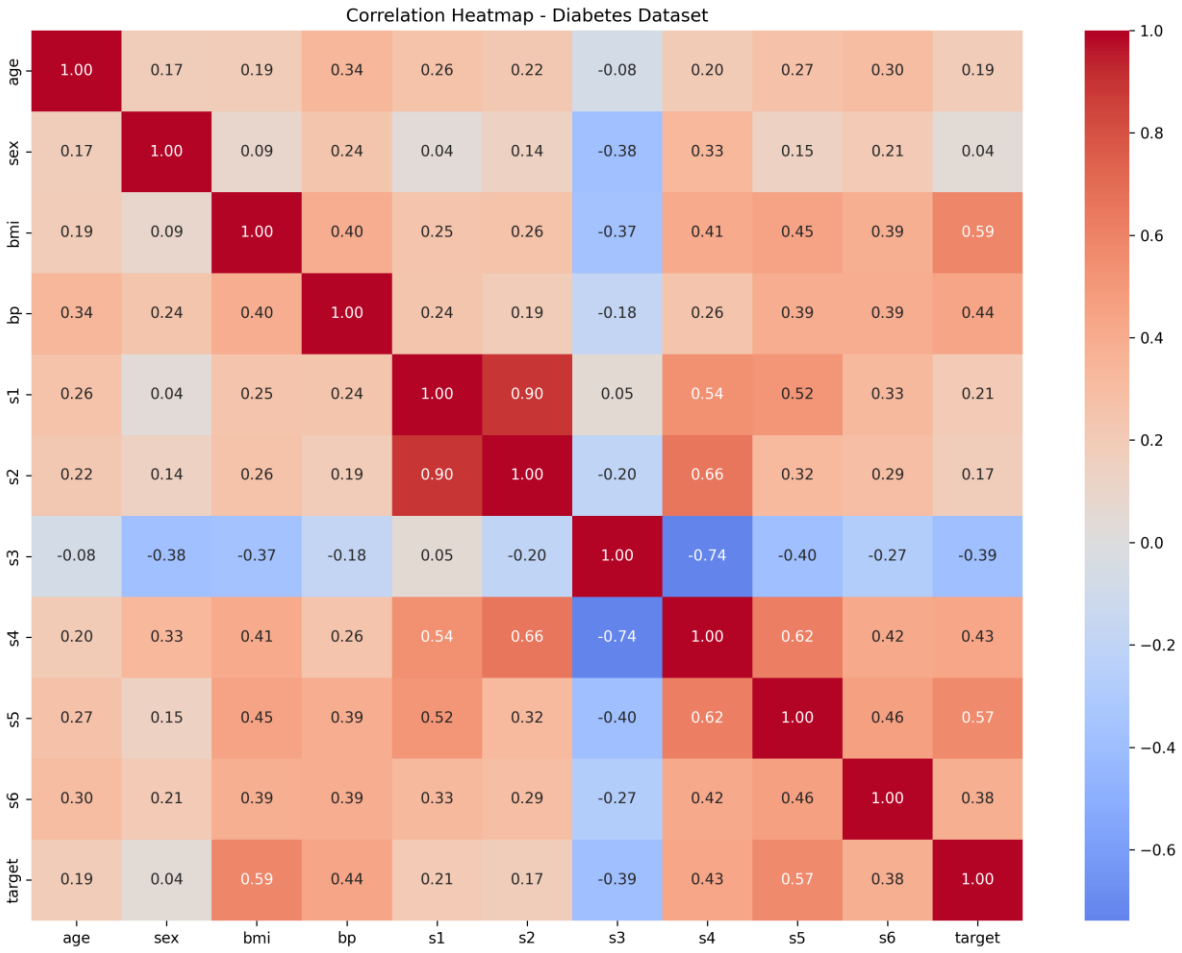


Figure 2: Correlation heatmap of dataset variables

6.5 BMI vs Target Scatter Plot

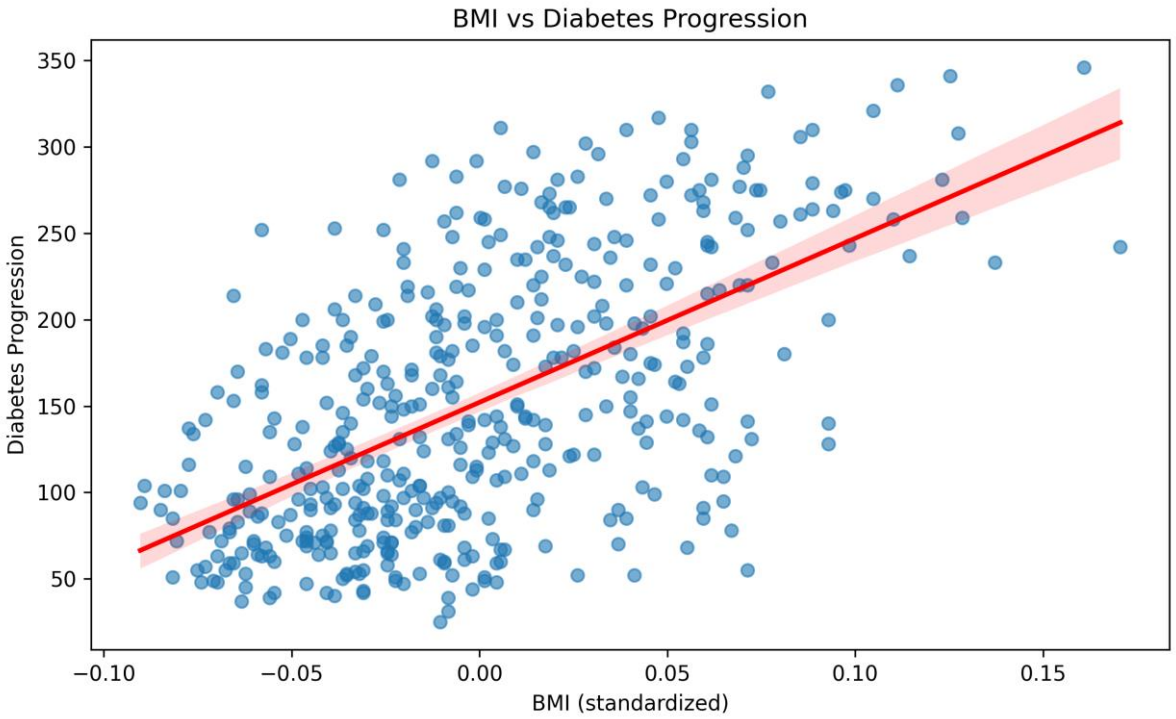


Figure 3: Relationship between BMI and target

7. Result

Correlation analysis was successfully performed on the Diabetes dataset. Among the ten independent variables, BMI showed the strongest absolute linear correlation with the target (r = 0.5865). The variables s5, bp and s4 also showed positive associations, while s3 showed a negative association.

8. Observations

- BMI has the strongest positive linear association with the target.
- The correlation coefficient for BMI is approximately 0.5865.
- The feature s5 has the second strongest positive correlation (approximately 0.5659).
- The feature s3 has a negative correlation of approximately -0.3948.
- Sex has a very weak linear correlation with the target (approximately 0.0431).
- Correlation indicates association and should not be interpreted as proof of causation.

9. Conclusion

The experiment successfully identified the association between the dependent and independent variables using Pearson correlation analysis. The bar chart and heatmap provided an overall view of the relationships, while the BMI-versus-target scatter plot gave a visual representation of the strongest association. The analysis shows that BMI has the strongest positive linear relationship with the target among the features considered.